

Schedule-Only Retries in Anonymous DHTs

When retry logic becomes a linkability channel

February 28, 2026 (archive v125)

Abstract

Retries, hedges, failover, and negotiation wrappers are ubiquitous in networked retrieval. Even when each individual attempt is secure (e.g., via an anonymous channel, an anonymous DHT lookup layer, or an oblivious relay), the *retry trace*—how many attempts were made, in what public schedule, and under which lane—is itself a high-bandwidth metadata channel that an active adversary can amplify via selective abort. This paper formalizes *Trace Privacy with Abort* (TPA) and introduces *Schedule-Only Retry Policies* (SRP): wrapper policies whose externally visible control flow depends only on a *public schedule*, not on hidden outcomes. We prove a lifting theorem: if each attempt satisfies privacy-with-abort (PWA) with advantage δ and the wrapper is ε -close to schedule-only, then wrapper-level TPA holds with advantage at most $T\delta + \varepsilon$, where T is the public slot budget. We instantiate SRP for committee contact in an anonymous DHT and provide a synthetic simulation showing how best-effort retries amplify advantage across epochs and how SRP arrests this growth. Stop-time padding and schedule optimization are treated as prior work in this series (e.g., [11, 14]).

Scope. We isolate and formalize *Schedule-Only Retries in Anonymous DHTs* as a reusable contract surface in the anonymity / Anonymous-DHT series.

Imports. We treat stop-time padding and the compiler/scheduling machinery as imported from the backbone posts.[11, 14] This operational paper specializes that machinery to the retry surface and to schedule-only deployments. Notation conventions for keys, state, transcripts, and transcript projections follow the series notation sheet (Synthesis 8).[44] The “equalization contract” viewpoint (Anonymity B) is an additional import: we treat the retry trace as one concrete transcript projection X whose skew must be budgeted and certified.[41]

Artifact policy. This archive is source-only; supporting artifacts (logs, traces, scripts, certificates, and verifier outputs) are available on request as a single bundle, committed by ABOM/OINL (Anonymity C) and governed by the series artifact policy (Synthesis 6).[40]

Exports. Exports a *schedule-only retry wrapper* specification: a canonical retry-trace projection X_{retry} for the equalization contract (Anonymity B).[41] plus a retry accountant compatible with odds-inflation budgeting, plus an *SRP plan* contract object (Synthesis 0) suitable for CPPC disclosure and change-control receipts,[39] plus a deployment checklist for downgrade surfaces and upgrade/fallback control-flow (transport privacy upgrades; canonical references in the related-work map, Synthesis 7)[45] and policy gates that fail closed when retry leakage regresses.

1 Introduction

Anonymous overlays and anonymous DHTs aim to hide *who contacts whom* during retrieval. However, real retrieval stacks nearly always include wrappers: timeouts, retries, hedged requests, multi-path probing, protocol negotiation, and gateway failover [18, 19]. Widely deployed RPC stacks make these wrappers explicit; for example, gRPC standardizes client retry and hedging behavior (including cancellation semantics) at the library level.[28] Even “mundane” connectivity algorithms explicitly recommend racing alternatives: for example, Happy Eyeballs (RFC 8305) improves responsiveness by attempting multiple address families using concurrency, i.e., a structured form of hedging/fallback [27]. These wrappers create a *trace*: a sequence of (attempt, wait, abort/accept, lane choice) events. An active adversary can *selectively abort* attempts to amplify information leaked in the trace, echoing selective denial-of-service attacks in anonymity systems [1, 2, 3].

More broadly, as encrypted transports hide payload contents, traffic analysis shifts to side-channel features such as timing, size, and control-flow metadata; recent work formalizes that such side-channel leakage can be information-theoretically inevitable under broad conditions.[33]

Goal. Design wrappers that preserve anonymity (and more broadly, metadata privacy) even in the presence of aborts, without forbidding practical performance tools like hedging and failover.

Contributions.

- We define TPA, a wrapper-level indistinguishability notion where the adversary observes retry traces and can induce aborts.
- We define SRP, a discipline for wrappers whose visible behavior is *schedule-only*.
- We prove a lifting theorem: composing up to T attempts that each satisfy PWA(advantage δ) under an SRPwrapper with deviation ε yields TPAwith advantage at most $T\delta + \varepsilon$.
- We give an SRPinstantiation for *committee contact in an anonymous DHT* and supply a synthetic simulation showing trace-amplification over epochs and its arrest under SRP.

Series positioning. This paper focuses on *retries/abort traces*. We refer to earlier series papers for stop-time padding, schedule compilation, and dual-certificate privacy accounting ([11, 12, 13, 14]).

2 Model and Notation

A *base attempt protocol* Π is invoked repeatedly by a *wrapper* W until W decides to stop. Each attempt emits an adversary-visible transcript and ends in an outcome in $\{\text{accept}, \text{abort}\}$. A *trace* records the wrapper’s externally visible *control-flow* events: attempt start times (or slots) and lane labels. Per-attempt transcripts (including whether a controlled recipient responds) are accounted for at the attempt layer via PWA; SRPconstrains only the wrapper’s visible control flow. We treat stop-time padding/termination-time hiding as prior work in this series (so that per-attempt completion times need not appear in the public trace).

A *public schedule* σ is a (possibly randomized) skeleton describing which slots are executed, with which lane labels, and with what timing class. We treat σ as public: it may be logged, audited, and used by adversaries. The design objective is that *secrets do not modulate externally visible traces beyond what σ allows*.

Figure omitted in source-only archive.
 (Plot/diagram and scripts available on request.)

Figure 1: A minimal SRP pipeline: a *trace contract* specifies a public schedule skeleton; compilation produces a wrapper that executes the skeleton and dummy-fills after success.

3 Security Definitions

We model two layers.

3.1 PWA: attempt-level privacy-with-abort (imported)

An attempt protocol Π is invoked in slots according to a public schedule σ . We assume Π comes with a composition-ready per-slot guarantee: for every executed slot and every public history consistent with σ (including adversarial abort/delay choices), the conditional distinguishing advantage between secrets is at most δ . This is the standard hypothesis used for sequential composition in Anonymity A and is exportable as a conditional per-step certificate (Synthesis 21).[42, 43]

3.2 TPA: wrapper-level trace privacy with abort

Fix a wrapper W and base attempt protocol Π . The adversary chooses two secrets (s_0, s_1) , observes the externally visible wrapper trace (slot indices, lane labels, and externally visible start/stop events), and may induce aborts/delays by controlling a subset of recipients/relays. Let $\text{Adv}_{W(\Pi)}^{\text{TPA}}$ denote the resulting distinguishing advantage (advantage conventions follow the archive’s contracts).[38] We target negligible Adv^{TPA} .

The gap between PWA and TPA is the wrapper: even if each attempt is safe, wrapper control flow may leak.

4 Schedule-Only Retry Policies (exorpdfstringSRPSRP)

Definition 1 (SRP(schedule-only)). *A wrapper policy W is SRP if, for every public schedule σ and any two secrets/outcome streams, the distribution of its externally visible trace conditioned on σ is identical. Equivalently: conditioned on σ , the trace is a function only of public randomness, not of secret-dependent success/failure.*

Operationally, SRP is achieved by fixing a per-epoch schedule and dummy-filling remaining slots after early success. A minimal trace-contract/compilation pipeline is shown in figure 1.

Approximate SRP. Real systems deviate (bugs, misconfiguration, shared pools). We quantify deviation as a statistical distance.

Definition 2 (Schedule-only deviation ε). *Fix σ . Let Trace_s denote the induced trace distribution when running $W(\Pi, s)$ with schedule σ . Define*

$$\varepsilon_\sigma(W) := \sup_{s_0, s_1} \text{TV}(\text{Trace}_{s_0}, \text{Trace}_{s_1}), \quad \varepsilon(W) := \sup_\sigma \varepsilon_\sigma(W).$$

W is perfect SRP iff $\varepsilon(W) = 0$.

Figure omitted in source-only archive.
 (Plot/diagram and scripts available on request.)

Figure 2: Illustration (from the research archive): selective abort can convert a weak per-epoch trace signal into near-certain inference by repeated epochs.

SRP as an equalization contract for the retry trace

Fix a public schedule σ and let X_{retry} denote the wrapper-visible trace (slot indices, lane labels, and any publicly logged schedule metadata). By definition, the schedule-only deviation $\varepsilon_\sigma(W)$ is the worst-case total-variation separation of the X_{retry} laws across secrets, conditioned on σ . Thus for any secrets s_0, s_1 and any measurable event E over traces,

$$\Pr[X_{\text{retry}} \in E \mid s_0, \sigma] \leq \Pr[X_{\text{retry}} \in E \mid s_1, \sigma] + \varepsilon_\sigma(W),$$

i.e., X_{retry} satisfies a pairwise $(0, \varepsilon_\sigma(W))$ equalization contract in the sense of Anonymity B.[41] Perfect SRP($\varepsilon = 0$) spends zero retry-trace budget; imperfect SRP exposes an explicit additive δ term that composes over epochs via the standard accountant.[41]

Operational note (monitoring). Deployments rarely inspect full traces; instead they commit to and monitor a small plan-defined feature family (conformance predicates, negotiation bits, coarse summaries). We treat *plan commitments*, transparency anchoring, and anytime-valid drift monitoring as imported modules: see Proof-Carrying Budgets (Boss Fight C) for certificate patterns, Release receipts for change-control binding, and Advantage Contracts (Eval 2) for anytime-valid monitoring recipes and scalable drift discovery.[36, 37, 38]

5 Why SRP is Close to Necessary

If a wrapper’s schedule depends on hidden outcomes, then an adversary can often amplify leakage by forcing aborts. This is the same qualitative mechanism as selective-DoS attacks [1, 2].

Proposition 1 (Amplification by selective abort (informal)). *Suppose W retries until success and the (slot-count or retry-count) trace is adversary-visible. If an adversary can cause aborts with probability depending on a hidden bit, then per-epoch trace distributions separate across the bit. Across independent epochs, likelihood-ratio accumulation drives distinguishing advantage toward $1/2$.*

6 Lifting: PWA+ SRP $\Rightarrow$ TPA

The key point is separation of concerns: if the wrapper’s trace is schedule-only, then all secret dependence must flow through the attempt transcripts, which are controlled by PWA.

Theorem 1 (Lifting bound). *Let Π satisfy attempt-level PWA with advantage δ per executed attempt slot, in the conditional/composition-ready sense (cf. Anonymity A or Synthesis 21).[42, 43] Let W be a wrapper with schedule-only deviation $\varepsilon(W)$, and suppose each epoch’s public schedule contains at most T attempt slots. Then for any adversary $\mathcal{A}$,*

$$\text{Adv}_{W(\Pi)}^{\text{TPA}}(\mathcal{A}) \leq T\delta + \varepsilon(W).$$

Here T counts the total number of executed attempt slots (including parallel/hedged slots if the schedule starts multiple attempts concurrently); wall-clock concurrency does not change the hybrid loss. In particular, perfect SRP wrappers ($\varepsilon = 0$) lift PWA to TPA with only a linear-in-slots loss from composing T attempts.

Proof sketch. Hybridize in two steps. First, hybrid over the at most T attempts and replace each attempt transcript with the PWA simulator; this changes the adversary’s view by at most $T\delta$. This step remains valid against adaptive abort strategies because δ is defined as a worst-case conditional bound over histories. Second, couple the remaining executions across secrets; the only remaining difference is the wrapper trace distribution, which differs by at most $\varepsilon(W)$ in total variation. $\square$

Lemma 1 (Product-TV bound). *Let View_0 and View_1 be the per-epoch adversary views, and suppose $\text{TV}(\text{View}_0, \text{View}_1) \leq \alpha$. If epochs are independent, then for E epochs,*

$$\text{TV}(\text{View}_0^{\otimes E}, \text{View}_1^{\otimes E}) \leq 1 - (1 - \alpha)^E \leq E\alpha.$$

Corollary 1 (Multi-epoch composition). *If the system runs for E independent epochs (fresh schedules, fresh coins), then*

$$\text{Adv}^{\text{TPA}} \leq 1 - (1 - (T\delta + \varepsilon(W)))^E \leq E(T\delta + \varepsilon(W)).$$

The first inequality is a sharper “product distance” bound; the second is the familiar union bound. Operationally, both highlight that even small deviations ε can matter at scale.

Dependence across epochs (brief caveat). The product-TV bound assumes independent epochs. Without independence, a standard hybrid/triangle argument still yields the safe bound $\text{TV}(\text{View}_0^{(1:E)}, \text{View}_1^{(1:E)}) \leq \sum_{i=1}^E \alpha_i$ when α_i bounds the per-epoch (possibly history-dependent) conditional distance. Thus the union bound remains valid even under adaptive network conditions; anytime-valid monitoring (Eval 2) does not require independence to flag drift.

7 How to Implement SRP (practical sketch)

We compile a best-effort retry policy into a schedule-only wrapper:

- **Choose a public schedule skeleton.** Examples include fixed slot budgets, lane schedules for negotiation/failover, or public-coin randomized stop-times.
- **Execute attempts on schedule.** Attempt completion can be recorded internally; visible behavior continues.
- **Dummy fill after success.** Remaining slots run cover traffic (or indistinguishable no-op work).

Tooling note. In practice, wrappers are implemented in complex control-flow languages (RPC middleware, service meshes, client libraries). Our artifact includes a small SRPlinter/repair pass and a compiler from *trace contracts* (public schedule skeletons) to wrappers that dummy-fill after success; the associated demo figures ([fig_srp_lint_demo.pdf](#), [fig_srp_compile_demo.pdf](#)) illustrate how secret- or outcome-dependent branching is detected and removed.

Figure omitted in source-only archive.
 (Plot/diagram and scripts available on request.)

Figure 3: Negotiation wrappers (upgrade-then-fallback) can create an amplifiable trace bit unless lane schedules are made SRP.

Folding in negotiation wrappers. Negotiation wrappers ("try privacy upgrade, then fall back") can create an amplifiable *lane bit* unless lane choices are fixed by a public schedule. This applies to upgrade/fallback mechanisms such as ECH, DDR, OHTTP, and ODoH; we treat their standards-level details (and the surrounding downgrade surfaces) as background and refer to the series related-work map (Synthesis 7) for canonical citations and context.[45] Operational proposals for authenticated ECH config distribution and rotation underscore that "upgrade then fall back" control flow must be treated as a first-class downgrade surface (and thus should be schedule-pinned).[9] To make negotiation wrappers SRP, treat the lane schedule as part of the public plan: select among a small set of pre-committed lane schedules using only pinned public signals (or public coins), then execute the schedule to completion with dummy fill after success. Outcome-dependent branching ("if upgrade fails then switch lanes") is prohibited; instead, commit to lane schedules (and their conformance predicates) as plan objects and audit them via the proof-carrying/receipt pipeline.[36, 37] figure 3 illustrates the risk.

8 Case Study: Committee Contact in an Anonymous DHT

Anonymous DHT lookups aim to hide initiator/target while contacting a sequence (or committee) of nodes. Recent proposals in other settings (e.g., blockchain overlays) similarly target initiator anonymity, query unlinkability, and target privacy; our wrapper-level argument applies to such designs whenever retries and negotiation are observable.[8] Prior work identifies lookup leakage attacks and builds dedicated anonymous lookup mechanisms[4, 5]; robust designs often rely on quorum/committee contact[20].

Our emphasis is orthogonal to attempt primitives: widely deployed Kademlia-family overlays (including IPFS/libp2p) use parallelism, timeouts, and retries to tolerate churn[6], parameterized by the number of closest peers k_{kad} returned/queried per step and by lookup concurrency α (e.g., the original Kademlia paper uses $\alpha = 3$), while the IPFS Kademlia DHT specification recommends $\alpha = 10$. [6, 10] Those wrappers are an unavoidable privacy surface: active disruption (eclipse/Sybil) can force retries[25, 29, 7]. Beyond overlay-level attacks, AS-level interception and BGP hijacking can censor IPFS content retrieval and provider discovery, inducing failures and retries at scale.[34] In IPFS, the Kademlia DHT specification explicitly permits early termination in several lookup modes, and recent work *reports* an active Sybil eclipse that exploits an early trigger of content-discovery termination in Kubo (the main IPFS implementation), yielding high denial rates in their evaluation.[10, 24] Meanwhile, complementary work targets attempt-level leakage in these systems (e.g., query obfuscation and PIR-style private queries)[21, 22]. Lightweight identifier-obfuscation proposals for IPFS (e.g., multi-hash schemes) further emphasize that partial defenses are easy to bypass via fallback behavior unless the wrapper's external trace is schedule-only.[31, 32]

Committee contact wrapper. Consider contacting a committee of size m per epoch and needing t successes. A best-effort strategy retries until it obtains t successful replies. This is efficient but leaks via the number and timing of attempts.

Parameter	Value
Success threshold t	3
SRP padded slots S	8
Lookup concurrency α (parallel evidence)	10
Corrupt-recipient fraction f	0.35
Selective abort (a_0, a_1)	(0.15, 0.85)
Honest failure / churn proxy	0.20
Schedule-deviation ε	0.02

Table 1: Simulation parameters for figure 4; defined in `evidence/dht_retry_simulation.py`. The choice of lookup concurrency α reflects the IPFS Kademlia DHT specification.[10]

Figure omitted in source-only archive.
(Plot/diagram and scripts available on request.)

Figure 4: Synthetic committee-contact simulation (script and CSV in `evidence/`): best-effort retries amplify advantage toward 1/2 across epochs; SRParrests amplification; small deviations accumulate slowly.

SRPinstantiation. Fix a public per-epoch schedule of S contact slots with a public lane map (e.g., slot indices assigned to a “primary” lane and a “backup” lane). For each slot, the wrapper invokes the attempt primitive once; the attempt primitive may internally choose which committee member to contact (e.g., via anonymous relay selection in Octopus-style lookups), so the wrapper trace reveals only slot/lane boundaries. After t successes, the wrapper continues to execute the schedule using dummies/no-ops. The adversary observes the same schedule regardless of success/failure, yielding TPAsubject to PWA.

9 Evaluation: amplification, arrest, and overhead

We include a small synthetic simulation calibrated to this committee-contact story (parameters in Table 1).

An adversary controls a fraction of recipients and selectively aborts attempts based on a hidden bit. Best-effort retries rapidly amplify distinguishing advantage with repeated epochs, while SRParrests this growth. figure 4 plots the advantage vs epochs.

Additional evidence (omitted for space). The archive includes small sweeps over corruption and churn parameters (CSVs + figures), plus a round-based parallelism variant using libp2p-style lookup concurrency ($\alpha = 10$) and an $\alpha \in \{3, 10\}$ sweep; all preserve the same qualitative conclusion.

External overhead vs internal latency. In an SRPwrapper, the *user-visible completion time* can match best-effort (success is detected internally), while *externally visible work* is padded to a schedule. Table 2 summarizes expected external attempt counts under the simulation parameters.

10 Related Work (selected)

Selective disruption/DoS lowers anonymity by forcing rebuilds/retries and biasing which traces occur [15, 1, 2, 23, 3]. Classic overlay/DHT security work on Sybil and eclipse attacks emphasizes

Scheme	$\mathbb{E}[\text{\#attempts} \mid s_0]$	$\mathbb{E}[\text{\#attempts} \mid s_1]$	leakage signal?
Best-effort	3.96	5.33	yes (means differ)
SRP(perfect, $S=8$)	8.00	8.00	no (schedule-only)
SRP(2% deviation, $S=8$)	7.92	7.95	small (accumulates)

Table 2: Per-epoch expected *externally visible* attempt counts (computed by `evidence/dht_retry_simulation.py`). Perfect SRP removes the per-epoch signal at the cost of fixed padding.

that adversaries can actively induce lookup failures or dominate neighborhoods, thereby creating retries as a controllable channel.[25, 26]

Anonymous DHT lookup mechanisms and analyses of lookup-related information leaks motivate strong attempt primitives [16, 17, 4, 5]. Our focus is orthogonal: *even with strong anonymous attempts*, wrapper retries and negotiation can reintroduce metadata leakage unless externally visible behavior is schedule-only.

At a higher level, SRP is an instance of data-oblivious control flow: like oblivious RAM (ORAM), it enforces that an adversary-visible schedule is independent of secret-dependent outcomes.[35]

Retry schedules in deployed protocols. Many protocols standardize externally visible retransmission and recovery schedules (e.g., QUIC loss recovery timers), emphasizing that the retry schedule is a common and frequently tuned surface.[30]

11 Limitations (brief)

SRP protects only the *externally visible retry trace*. It does not, by itself, hide total volume/rate, long-lived identifiers, or other side channels in the attempt primitive. Nor does it prevent denial-of-service; rather, it prevents an attacker from turning induced failures into an amplifiable *trace* signal. Finally, SRP assumes that schedule selection is a public, auditable function of pinned public inputs (or public coins); if a deployment makes schedule choice depend on secret state, it reintroduces the very channel SRP is meant to close.

12 Conclusion

Retries are not “just engineering”: they are a first-class privacy surface. SRP offers a clean discipline for abort-safe wrappers and a path to compositional theorems. Operationally, SRP works best when the schedule skeleton and conformance predicates are committed as plan objects and monitored for drift with anytime-valid evidence; those publication mechanics are owned by Boss Fight C, Release receipts, and Eval 2.[36, 37, 38]

13 Takeaways

- **Retries are a metadata channel:** the retry trace (slot count, lane choice, scheduling) can leak far more than a single attempt.
- **Abort amplifies:** selective abort turns weak per-epoch trace signals into near-certain inference across repeated epochs.

- **Schedule-only is the right discipline:** fix a public schedule and dummy-fill after early success; then wrapper-level trace privacy composes with the attempt primitive.

Artifacts (available on request)

This paper is derived from an internal research archive that includes: `tools/` for linting/compiling SRPolicies and `evidence/` scripts reproducing figure 4, Table 2, the control-plane lane-bit amplification figure, and the TV composition bound plot. We publish the paper text and figures; additional artifacts are available on request.

References

- [1] Nikita Borisov and George Danezis and Prateek Mittal. Denial of Service or Denial of Security?. ACM Conference on Computer and Communications Security (CCS), 2007. <https://www.freehaven.net/anonbib/cache/ccs07-doa.pdf>
- [2] Anupam Das and Nikita Borisov. Securing Tor Tunnels under the Selective-DoS Attack. Financial Cryptography and Data Security (FC), 2013. <https://arxiv.org/abs/1107.3863>
- [3] Rob Jansen and Tavish Vaidya and Micah Sherr. Point Break: A Study of Bandwidth Denial-of-Service Attacks against Tor. 28th USENIX Security Symposium (USENIX Security 19), 2019. https://www.usenix.org/system/files/sec19fall_jansen_prepub.pdf
- [4] Qiyang Wang and Nikita Borisov. In Search of an Anonymous and Secure Lookup: Attacks on Structured Peer-to-Peer Anonymous Communication Systems. ACM Conference on Computer and Communications Security (CCS), 2010. <https://www.freehaven.net/anonbib/cache/ccs10-lookup.pdf>
- [5] Qiyang Wang and Nikita Borisov. Octopus: A Secure and Anonymous DHT Lookup. IEEE International Conference on Distributed Computing Systems (ICDCS), 2012. <https://arxiv.org/abs/1203.2668>
- [6] Petar Maymounkov and David Mazieres. Kademlia: A Peer-to-Peer Information System Based on the XOR Metric. International Workshop on Peer-to-Peer Systems (IPTPS), 2002. <https://www.scs.stanford.edu/~dm/home/papers/kpos.pdf>
- [7] Srivatsan Sridhar and Onur Ascigil and Navin Keizer and François Genon and Sébastien Pierre and Yannis Psaras and Etienne Rivière and Michał Król. Content Censorship in the InterPlanetary File System. Network and Distributed System Security Symposium (NDSS), 2024. DOI: 10.14722/ndss.2024.23153. <https://www.ndss-symposium.org/wp-content/uploads/2024-153-paper.pdf>
- [8] Lewis Tseng and Lokesh Ambarapu and Moayad Alosaily. NC-DHT: a Robust and Anonymous DHT for Blockchain Systems. International Conference on Blockchain Computing and Applications (BCCA), 394–399, 2024. DOI: 10.1109/BCCA62388.2024.10844445. <https://doi.org/10.1109/BCCA62388.2024.10844445>
- [9] Nick Sullivan and Dennis Jackson. Authenticated ECH Config Distribution and Rotation (Updates). Internet-Draft, `draft-sullivan-tls-signed-ech-updates-00`, 2025. Highlights operational needs around ECH config updates/rotation; relevant to failure-handling wrappers and downgrade surfaces. Accessed 2026-02-22. <https://datatracker.ietf.org/doc/draft-sullivan-tls-signed-ech-updates/>
- [10] IPFS/Libp2p. IPFS Kademlia DHT Specification. Online specification, 2025. Describes lookup and early-termination conditions; accessed 2026-02-22. <https://specs.ipfs.tech/routing/kad-dht/>
- [11] Anonymous. Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups. Series preprint, 2026.

- [12] Anonymous. Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding. Series preprint, 2026.
- [13] Anonymous. Knapsack, Transport, and Dual Certificates for TV Privacy. Series preprint, 2026.
- [14] Anonymous. Stop-Time Padding Addendum (unique-only): Max-Mixture Separations and Tail-Sign Deadline Normal Forms. Series preprint, 2026.
- [15] Bauer, Kevin and McCoy, Damon and Grunwald, Dirk and Kohno, Tadayoshi and Sicker, Douglas. Low-Resource Routing Attacks Against Tor. Proceedings of the ACM Workshop on Privacy in the Electronic Society (WPES), 2007. <https://www.freehaven.net/anonbib/cache/bauer%3Awpes2007.pdf>
- [16] Nambiar, Anand and Wright, Matthew and Levine, Brian N.. Salsa: A Structured Approach to Large-Scale Anonymity. Proceedings of the ACM Symposium on Operating Systems Principles (SOSP), 2006. <https://www.freehaven.net/anonbib/cache/Salsa.pdf>
- [17] Prateek Mittal and Nikita Borisov. Information Leaks in Structured Peer-to-Peer Anonymous Communication Systems. *ACM Transactions on Information and System Security (TISSEC)*, 15(1), Article 5, 2012. DOI: 10.1145/2133375.2133380. <https://dl.acm.org/doi/10.1145/2133375.2133380> (Author PDF: <https://www.princeton.edu/~pmittal/publications/information-leaks-tissec.pdf>)
- [18] Jeffrey Dean and Luiz André Barroso. The Tail at Scale. *Communications of the ACM*, 56(2), 74–80, 2013. DOI: 10.1145/2408776.2408794. <https://research.google/pubs/the-tail-at-scale/>
- [19] Mia Primorac and Katerina Argyraki and Edouard Bugnion. When to Hedge in Interactive Services. 18th USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2021. https://www.usenix.org/system/files/nsdi21spring-primorac_0.pdf
- [20] Michael Backes and Ian Goldberg and Aniket Kate and Esfandiar Mohammadi. Adding Query Privacy to Robust DHTs. *Applied Cryptography and Network Security (ACNS)*, 2012. <https://cypherpunks.ca/~iang/pubs/anonymous-p2p.pdf>
- [21] Erik Daniel and Guillaume Michel and Florian Tschorsch. Improving Kademlia Lookup Privacy through Query Obfuscation. Proceedings of the 40th ACM/SIGAPP Symposium on Applied Computing (SAC '25), 1345–1352, 2025. DOI: 10.1145/3672608.3707703. <https://guillaume.michel.id/docs/kademlia-query-obfuscation.pdf>
- [22] Miti Mazmudar and Shannon Veitch and Rasoul Akhavan Mahdavi. Peer2PIR: Private Queries for IPFS. IEEE Symposium on Security and Privacy (SP), 4438–4456, 2025. DOI: 10.1109/SP61157.2025.00231. Open-access copy available via ETH Research Collection (see <https://www.research-collection.ethz.ch/handle/20.500.11850/733500>). Accessed 2026-02-22. <https://doi.org/10.1109/SP61157.2025.00231>
- [23] Norman Danner and Danny Krizanc and Marc Liberatore. Detecting Denial-of-Service Attacks in Tor. *Financial Cryptography and Data Security (FC)*, 2009. https://fc09.ifca.ai/papers/43_Detecting_dos_in_tor.pdf
- [24] Netto, V. H. M. and Cholez, Thibault and Ignat, Claudia-Lavinia. Active Sybil Attack and Efficient Defense Strategy in IPFS DHT. arXiv:2505.01139, 2025. <https://arxiv.org/abs/2505.01139>
- [25] Singh, Atul and Ngan, Tsuen-Wan “Johnny” and Druschel, Peter and Wallach, Dan S.. Eclipse Attacks on Overlay Networks: Threats and Defenses. Proceedings of IEEE INFOCOM, 2006. DOI: 10.1109/INFOCOM.2006.231. <https://www.cs.rice.edu/~dwallach/pub/eclipse-infocom06.pdf>
- [26] Baumgart, Ingmar and Mies, Sebastian. S/Kademlia: A Practicable Approach Towards Secure Key-Based Routing. International Conference on Parallel and Distributed Systems (ICPADS), 1–8, 2007. DOI: 10.1109/ICPADS.2007.4447808. https://telematics.tm.kit.edu/publications/Files/267/SKademlia_2007.pdf

- [27] Schinazi, David and Pauly, Tommy and Jaju, Nidhi and Ishibashi, Kenichi. RFC 8305: Happy Eyeballs Version 2: Better Connectivity Using Concurrency. IETF Proposed Standard, 2017. <https://datatracker.ietf.org/doc/html/rfc8305>
- [28] gRPC Project. Client Retries and Hedging in gRPC (A6 proposal). Design proposal, `A6-client-retries.md`, 2024. Documents retry and hedging semantics (including cancellation/deadlines) in the widely deployed gRPC client library. Accessed 2026-02-22. <https://github.com/grpc/proposal/blob/master/A6-client-retries.md>
- [29] Zhang, Ruirui and Hu, Y. Charlie. Making Eclipse Attacks Computationally Infeasible in Large-Scale DHTs. Proceedings of the 3rd International Workshop on Peer-to-Peer Systems (IPTPS), 2011. <https://nirenzang.github.io/assets/Making-Eclipse-Attacks-Computationally-Infeasible-in-Large-Scale-DHTs.pdf>
- [30] Iyengar, J. and Swett, I.. QUIC Loss Detection and Congestion Control. IETF, 2021. <https://www.rfc-editor.org/rfc/rfc9002.html>
- [31] Thomas Katsantas and Yannis Thomas and Christos Karapapas and George Xylomenos. Enhancing IPFS Privacy Through Triple Hashing. 2024 IEEE Symposium on Computers and Communications (ISCC), 1–6, 2024. DOI: 10.1109/ISCC61673.2024.10733729. <https://mm.aueb.gr/publications/5b9cdb00-fa7d-4e93-b2d0-31f509116371.pdf>
- [32] Erik Daniel and Marcel Ebert and Florian Tschorsch. Improving Bitswap Privacy with Forwarding and Source Obfuscation. arXiv:2307.03480, 2023. <https://arxiv.org/abs/2307.03480>
- [33] Liu, Guangjie and Chen, Guang and Liu, Weiwei. The Inevitability of Side-Channel Leakage in Encrypted Traffic. arXiv:2602.14055, 2026. Submitted 2026-02-15; provides an information-theoretic argument that encrypted-traffic side channels can be unavoidable under broad conditions. Accessed 2026-02-22. <https://arxiv.org/abs/2602.14055>
- [34] Matter, Jan and Tran, Muoi. Network-level Censorship Attacks in the InterPlanetary File System. arXiv:2509.06626, 2025. Analyzes AS-level interception and BGP hijacking to censor IPFS content/provider discovery; relevant as a realistic active-failure source that can induce retries. Accessed 2026-02-22. <https://arxiv.org/abs/2509.06626>
- [35] Goldreich, Oded and Ostrovsky, Rafail. Software Protection and Simulation on Oblivious RAMs. Journal of the ACM, 43(3), 431–473, 1996. DOI: 10.1145/233551.233553. <https://dl.acm.org/doi/10.1145/233551.233553>
- [36] Anonymous. Proof-Carrying Budgets for Anonymous DHTs: Dual Certificates and Tiny Verifiers for Padding, Delegation, and Congestion Contracts. Boss Fight series Paper C, 2026. (This archive.)
- [37] Anonymous. Privacy as a Release Property: Proof-Carrying Anonymity Receipts for Anonymous DHTs. Release and destination Paper A, 2026. (This archive.)
- [38] Anonymous. Advantage Contracts for Anonymous DHT Evaluations: Stealth Audits, Leakage Discovery at Scale, and Practical Padding Filters. Evaluation series Paper 2, 2026. (This archive.)
- [39] Anonymous. Anonymity Contract Objects for Anonymous DHTs: A Practical Interface for Proof-Carrying Claims. Series synthesis note (Synthesis 0), 2026. (This archive.)
- [40] Anonymous. Artifact policy and supporting materials for the anonymity/AnonDHT series. 2026. See synthesis paper: Artifact policy (this archive). Artifacts, tooling, certificates, and scripts are available on request as a single archive.
- [41] Anonymous. Anonymous DHT Profiling and the Equalization Contract. Anonymity series Paper B, 2026. (This archive.)

- [42] Anonymous. Odds Inflation, Privacy Loss, and Endpoint Anonymity Bounds. Anonymity series Paper A, 2026. (This archive.)
- [43] Anonymous. Conditional Per-Step Budget Certificates. Series synthesis note (Synthesis 21), 2026. (This archive.)
- [44] Anonymous. Notation and Minimal Model for the One-True-Archive. Series synthesis note (Synthesis 8), 2026. (This archive.)
- [45] Anonymous. Related Work Map: Anonymous DHTs, Query Privacy, and Deployed Routing Ecosystems. Series synthesis note (Synthesis 7), 2026. (This archive.)